

UNIVERSITY OF CLAUDE · FREE RESOURCE MAP

Resource Map

CSCE 650 Virtual Reality — module-by-module mapping onto free university courseware

HOW TO USE THIS

This is your lecture hall. For each module: watch or read the mapped external material first, then work the module slides and notes, which are written to assume you have seen it.

Where a specific lecture number is given, that is the one that covers this module. Where a book chapter is given, read that chapter, not the whole book.

Everything here is free. Resources remain under their own licences — link to them, do not redistribute them.

Primary sources for the whole course

Resource	Role in this course
LaValle — Virtual Reality (free book and Illinois lectures)	The primary text, free and complete. Unusually strong on perception and on the mathematics of tracking, written by someone who worked on the Oculus tracking system.
Stanford EE267 — Virtual Reality, build your own HMD (free)	Course notes, slides, and assignments that build a headset from parts. The optics and display material is the best free treatment of Modules 03 and 04.
Oculus/Meta Developer Documentation — performance and comfort (free)	The practical engineering: frame budgets, timewarp, guardian, and the comfort guidelines. Vendor documentation, and genuinely good on the hard numbers.
Unreal Engine XR documentation (free)	What you will actually build in. Read the performance and stereo-rendering sections before Module 10.
Michael Abrash — blog archive and talks (free)	The latency and display analysis that shaped modern VR. 'Latency — the sine qua non of AR and VR' is required reading for Module 06.
Jerald — The VR Book (chapters and talks free)	The human-factors and design side, which the engineering sources underweight. Good for Modules 07, 08, and 13.

Module-by-module map

Module 01 — Presence, and What VR Must Deliver

Resource	What it covers for this module
LaValle — Virtual Reality, Chapters 1 and 12 (free)	The definition of presence, the history, and the evaluation chapter. Chapter 12 is the empirical-method argument of §3, properly developed.
Michael Abrash — What VR Could, Should, and Almost Certainly Will Be (free talk)	The clearest statement of why the requirements are what they are, from the period when they were being discovered.
Meta — VR design guidelines and comfort documentation (free)	The practical rules, with the numbers. Read the comfort section before Module 07.
Slater — Place illusion and plausibility (free)	The research framing of presence as two separable illusions — being there, and the events being real. More precise than the usual treatment.

Module 02 — Human Vision for VR

Resource	What it covers for this module
LaValle — Virtual Reality, Chapters 4 and 6 (free)	Light, optics, and the visual physiology of this module, with the numbers and their sources.
Stanford EE267 — human visual system and stereo lectures (free)	The acuity, depth cue, and vergence-accommodation material, with demonstrations you can run.
Hoffman et al. — Vergence-accommodation conflicts hinder visual performance and cause visual fatigue (free)	The study that quantified §2's conflict. The measurements are what justify the 0.5–2 m rule.
Abrash — Down the VR rabbit hole: fixing judder (free)	Persistence, judder, and why low persistence mattered so much. The best explanation of §3 anywhere.

Module 03 — Stereo Rendering

Resource	What it covers for this module
LaValle — Virtual Reality, Chapter 7 (free)	Visual rendering for VR, including the off-axis projection derivation of §1.
Stanford EE267 — stereo rendering lecture and assignment (free)	Builds the asymmetric frustum from scratch. Do the assignment; it makes the geometry concrete.
OpenXR specification — view configuration and projection (free)	What the runtime actually supplies, and in what coordinate conventions. The authority for §3's instruction.
Paul Bourke — calculating stereo pairs (free)	The clearest short explanation of why toe-in is wrong, with diagrams.

Module 04 — Display Optics and Distortion

Resource	What it covers for this module
Stanford EE267 — optics and distortion lectures and assignment (free)	Builds the distortion correction from the lens equation and has you implement it. The best free treatment of this module.
LaValle — Virtual Reality, Chapter 4 (free)	Light and optics, including the aberrations and why Fresnel lenses behave as they do.
Oculus — asynchronous timewarp and the compositor (free developer blog archive)	Why the compositor owns distortion, written when the decision was being made.
OpenXR — visibility mask extension (free)	The hidden-area mesh of §3. Short specification, immediate benefit.

Module 05 — Tracking

Resource	What it covers for this module
LaValle — Virtual Reality, Chapter 9 (free)	Tracking in full: IMU models, drift, filtering, and the mathematics of fusion. Written by someone who built the Oculus tracker, and the best free source for this module.
Stanford EE267 — IMU and pose tracking assignment (free)	Implement a complementary filter on real IMU data. The drift is far more convincing when you have watched it accumulate.
Madgwick — An efficient orientation filter for IMUs (free)	A widely used practical fusion filter, with source. A good implementation target.
OpenXR — space and pose prediction (free)	The predicted-display-time API of §3, and the pose validity flags you need for tracking loss.

Module 06 — Latency

Resource	What it covers for this module
Abrash — Latency: the sine qua non of AR and VR (free)	Required reading for this module. The clearest account of why latency matters and what the threshold means.
LaValle — Virtual Reality, Chapter 7.4 (free)	Timewarp, prediction, and the latency budget, with the mathematics.
Van Waveren — The Asynchronous Time Warp for Virtual Reality on Consumer Hardware (free)	The ATW paper of §2, including the GPU pre-emption problem that made it hard.
Meta — Asynchronous Spacewarp documentation (free)	What spacewarp needs from your application, and the artefacts to expect.

Module 07 — Cybersickness

Resource	What it covers for this module
LaValle — Virtual Reality, Chapter 10 (free)	VR sickness: the competing theories, the evidence, and the design implications. The best free treatment.
Kennedy et al. — Simulator Sickness Questionnaire (free, widely reproduced)	The standard instrument of §2. Use it rather than inventing your own scale.
Meta — VR comfort and locomotion guidelines (free)	The practical mitigations with the vendor's own testing behind them.
Rebenitsch & Owen — Review on cybersickness in applications and visual displays (free)	A survey of the evidence for §3's mitigations, including which ones replicate and which do not.

Module 08 — Locomotion

Resource	What it covers for this module
LaValle — Virtual Reality, Chapter 10 (free)	Locomotion and its relationship to sickness, with the research framing.
Steinicke et al. — Estimation of Detection Thresholds for Redirected Walking (free)	Where §4's threshold numbers come from. The experimental design is worth reading on its own.
Suma et al. — Impossible Spaces and Change Blindness Redirection (free)	The two most practical redirection techniques for small rooms.
Meta — locomotion design guidelines (free)	The practical comfort rules for each scheme, with vendor testing behind them.

Module 09 — Interaction and Input

Resource	What it covers for this module
LaValle — Virtual Reality, Chapter 10.2 (free)	Interaction mechanics and the research framing of manipulation in VR.
Meta — interaction and hand tracking design guidelines (free)	The practical rules, including hit volumes, gesture reliability, and the feedback recommendations of §3.
Casiez et al. — 1€ Filter (free, with code)	The input filter of §1. A page of code, immediately useful, and it works well.
Bowman et al. — 3D User Interfaces: Theory and Practice	The standard text on 3D interaction. Library, not free, and the selection and manipulation taxonomy is worth knowing.

Module 10 — Rendering for VR

Resource	What it covers for this module
Unreal Engine — XR performance and rendering documentation (free)	Instanced stereo, multiview, and the engine's VR rendering path. What you will actually configure.

Resource	What it covers for this module
Meta — fixed foveated rendering and performance guidelines (free)	Fixed foveation on mobile hardware, with the measured savings of §2.
Patney et al. — Towards Foveated Rendering for Gaze-Tracked Virtual Reality (free)	The perceptual basis of foveated rendering and the saccade constraint in §2.
NVIDIA — Variable Rate Shading documentation (free)	VRS in practice, including the shading-rate image interface.

Module 11 — Spatial Audio

Resource	What it covers for this module
Meta — Audio design and spatialisation documentation (free)	Practical VR audio: HRTF spatialisation, reverb zones, occlusion, and the latency requirements of §3.
Steam Audio documentation (free, with SDK)	Geometry-based early reflections and occlusion — §2's implementation, free and usable in both major engines.
Begault — 3-D Sound for Virtual Reality and Multimedia (NASA, free)	The foundational text on binaural audio and HRTFs, free from NASA. Thorough on §1.
SOFA HRTF databases (free)	Measured HRTF sets in a standard format, including several public databases. Useful for experimenting with individual variation.

Module 12 — Augmented Reality and Passthrough

Resource	What it covers for this module
Azuma — A Survey of Augmented Reality (free)	The foundational survey. The registration analysis of §2 is clearer here than in most modern treatments.
Meta — Passthrough and Mixed Reality documentation (free)	The current practice: passthrough APIs, scene understanding, anchors, and depth.
Apple — visionOS human interface guidelines (free)	The most carefully considered published thinking on §3's social and environmental design questions, whatever you make of the device.
Google — ARCore lighting estimation documentation (free)	Practical lighting estimation, including environment map generation from a camera feed.

Module 13 — Designing and Evaluating VR

Resource	What it covers for this module
LaValle — Virtual Reality, Chapter 12 (free)	Evaluation methodology, the standard instruments, and experimental design for perceptual claims.

Resource	What it covers for this module
Meta — VR design and accessibility guidelines (free)	The convention set of §1 and a growing accessibility section.
XR Access — accessibility resources (free)	The most active community work on §1's accessibility material, including guidelines and user research.
Nielsen — Why You Only Need to Test with 5 Users (free)	The basis for §2's claim about six participants. Short, and the argument generalises to VR.
Slater & Sanchez-Vives — Enhancing Our Lives with Immersive Virtual Reality (free)	A survey of what the medium can do, including the ethical questions in §3, from researchers who have studied them.